

Group Representation Theory

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1 Representations

Definition 1.0.1. A **representation** of a group G is a pair (V, ρ) where V is a vector space and $\rho : G \rightarrow GL(V)$ is a group homomorphism. So group action $G \curvearrowright V$ by invertible linear transformations. Much like with group actions, a representation is called **faithful** if ρ is injective.

Definition 1.0.2. For a group G , a homomorphism of representations $T : (V, \rho) \rightarrow (V', \rho')$ is a linear map $T : V \rightarrow V'$ such that for all $T \circ \rho(g) = \rho'(g) \circ T$ for all $g \in G$.

If we think about this categorically, then the category of k -representations of a group G is $\mathbf{Vect}_k^G$.

From now on, unless specified otherwise, all representations will be of a finite group G on a finite dimensional complex vector space. This lets us identify $GL(V)$ with $GL_n(\mathbb{C})$, the normal dictionary between linear maps and matrices from linear algebra applies everywhere.

Lemma 1.0.3. Two finite-dimensional representations (V, ρ) and (V', ρ') are isomorphic iff $\dim(V) = \dim(V')$ and there exists bases B and B' respectively such that there exists some $P \in GL_n(\mathbb{C})$ such that

$$[\rho'(g)]_{B'} = P^{-1}[\rho(g)]_B P$$

for all $g \in G$. In this case, the **matrix representations** are called **equivalent**.

We can now consider a few examples of representations.

For any group G , we can consider the one-dimensional representation $\rho(g) = \text{id}_{\mathbb{C}}$ for all g , this is called the **trivial representation**.

If ζ is a n th root of unit, then $C_n = \langle g \rangle$ has a one-dimensional representation by multiplication by $\rho(g^k)(z) = \zeta^k z$ for all $k \in \mathbb{Z}$, when $n = 1$ this recovers the trivial representation.

Let $G = S_n$, and let each $\sigma \in S_n$ act as the corresponding matrix of a basis B , i.e. $\rho(\sigma)(b_i) = b_{\sigma(i)}$ for all $b_i \in B$.

If $G = S_n$, then there is a one-dimensional representation with $\rho(g) = (\text{sign}(g))$. This is equal to applying the determinant to the permutation representation.

Let $G = D_{2n}$, this has natural two-dimensional representation on $\mathbb{R}^2$ sending each g to the associated transformation.

For any group action $G \curvearrowright X$ on a set X , there is an associated action of G on $\mathbb{C}[X]$, when we take $X = G$, and the action is by left multiplication, then the action of G on the $\mathbb{C}$ -algebra $\mathbb{C}[G]$ is called the **regular representation**. For any representation (V, ρ) , the natural map:

$$\phi : \text{Hom}_G(\mathbb{C}[G], V) \rightarrow V \quad \phi(T) = T(e)$$

is a G -linear isomorphism, with inverse $\phi^{-1}(v)(g) = \rho(g)v$.

1.1 Subrepresentations

Definition 1.1.1. A **subrepresentation** of a representation (V, ρ) is a subspace $W \leq V$ such that $\rho(g)(W) \subseteq W$ for all $g \in G$, we also call W a G -invariant subspace. Subrepresentations are called **proper** and **nonzero** if W satisfies the respective condition from linear algebra.

When W is finite dimensional, as $\rho(g)|_W \in GL(W)$ is a bijection, we in fact have $\rho(g)(W) = W$. When W is infinite dimensional this need not be true, only when $\rho(g)(W) \subseteq W$ for all $g \in G$ do we get $\rho(g)(W) = W$ for all $g \in G$.

Definition 1.1.2. A representation (V, ρ) is **irreducible** (or **simple**) if it is nonzero, and it does not admit any proper nonzero subrepresentations. Otherwise, it is called **reducible**.

Proposition 1.1.3. If G is a finite group with (V, ρ) an irreducible representation, then V is finite dimensional.

Proof. Let $v \in V$ be nonzero, and write W for the span of the $\rho(g)(v)$ for all $g \in G$, then this is a subrepresentation of V , as for all $h \in G$:

$$\rho(h) \sum_{g \in G} a_g \rho(g)v = \sum_{g \in G} a_g \rho(hg)v \in W$$

for all $a_g \in \mathbb{C}$. As v is nonzero, so is W , so by irreducibility of (V, ρ) , $V = W$, finite dimensional. $\square$

For an alternate proof, consider the map $\mathbb{C}[G] \rightarrow V$ induced by some nonzero $v \in V$, the image of this is a finite dimensional non-zero subrepresentation.

Proposition 1.1.4. If $T : (V, \rho) \rightarrow (V', \rho')$ is a homomorphism of representations, then $\ker(T)$ and $\text{im}(T)$ are subrepresentations of V and V' respectively.

Proof. $T(v) = 0 \implies T(gv) = g(Tv) = 0$ for all $g \in G$, $gT(v) = T(gv) \in \text{im}(T)$ for all $g \in G$. $\square$

Definition 1.1.5. A **quotient representation** of a representation (V, ρ_V) by a subrepresentation (W, ρ_W) is the vector space V/W with action $\rho(g)(v + W) = \rho(g)v + W$.

Proposition 1.1.6. For $T : (V, \rho) \rightarrow (V', \rho')$ is a homomorphism of representations, there is an isomorphism of representations:

$$(V, \rho) / \ker(T) \cong \text{im}(T)$$

Proof. We just have to show the map from the first isomorphism theorem $\bar{T}$ is G -linear. For all $g \in G$ and $v \in V$:

$$\bar{T} \circ \rho(g)(v + \ker(T)) = T \circ \rho(g)(v) = \rho(g)T(v) = \rho(g)\bar{T}(v + \ker(T)) \quad \square$$

Recall the following fact from linear algebra:

Lemma 1.1.7. For $T \in \text{End}(V)$ with $T^2 = T$, there is the direct sum decomposition $V = \ker(T) \oplus \text{im}(T)$.

Proof. Send $v \in V$ to $(v - T(v), T(v))$, this is immediately linear and bijective. $\square$

In this case, we call T a **projection operator**.

Corollary 1.1.8. If $T \in \text{End}(V)$ is a G -linear projection operator, then the direct sum of the previous lemma is a direct sum of subrepresentations.

Definition 1.1.9. A nonzero representation is **decomposable** if it can be written as a direct sum of two proper nonzero subrepresentations, it is called **indecomposable** otherwise.

A representation that is a direct sum of irreducible subrepresentations is called **semisimple**.

Indecomposable representations are semisimple iff they are irreducible. There are still examples of indecomposable but not irreducible representations:

Let $G = \mathbb{Z}$ and consider the two-dimensional representation:

$$\rho(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

This has the x -axis as a subrepresentation, but this is the only common one-dimensional eigen-space of $\rho(G)$, so there is not a full decomposition into irreducible subrepresentations.

1.2 Maschke's theorem

Definition 1.2.1. For (V, ρ) a representation, and $W \leq V$ a subrepresentation, a **complementary subrepresentation** is a subrepresentation $U \leq V$ such that $V = W \oplus U$.

Theorem 1.2.2 (Maschke). Let (V, ρ) be a finite dimensional representation of a finite group G . For any subrepresentation $W \leq V$, there exists a complementary representation.

Proof. We want to construct a projection operator with W as its image. By linear algebra, we can find a U such that $V = W \oplus U$, and the map $T(v) = T(w + u) = w$ is a good candidate. Unfortunately, this isn't necessarily G -linear, so we have to modify it, have $\tilde{T} : V \rightarrow V$ define as:

$$\tilde{T} = |G|^{-1} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1}$$

we first have to show this linear map is G -linear, for all $h \in G$ consider:

$$\tilde{T} \circ \rho(h) = |G|^{-1} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1} \rho(h) = |G|^{-1} \sum_{g' = h^{-1}g \in G} \rho(h) \rho(g') \circ T \circ \rho(g')^{-1} = \rho(h) \circ \tilde{T}$$

so $\tilde{T}$ is G -linear. To prove it is a projection operator we must show $\tilde{T}|_W = \text{id}$, for all $w \in W$:

$$\tilde{T}(w) = |G|^{-1} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1}(w) = |G|^{-1} \sum_{g \in G} \rho(g) \rho(g)^{-1}(w) = |G|^{-1} \sum_{g \in G} w = w$$

Finally, we must show $\text{im}(\tilde{T}) = W$, by the previous statement we know $\text{im}(\tilde{T}) \subseteq W$, but as T is a projection operator on W and W is G -invariant, we also have $W \subseteq \text{im}(\tilde{T})$. $\square$

We can now do induction on the dimension of the irreducible subspace we consider to get:

Corollary 1.2.3. Any finite dimensional representation of a finite group can be written as a direct sum of irreducible subrepresentations. i.e. V is semisimple.

Corollary 1.2.4. Finite dimensional indecomposable representations of finite groups are irreducible.

This doesn't actually require the vector space to be finite dimensional. Note, we haven't said anything about this decomposition being unique, there are examples where uniqueness does and doesn't hold:

If $G = 1$, then representations are the same as vector spaces, so the only irreducible representation of G is $\mathbb{C}$. For any bigger representation, the decomposition of is exactly the decomposition of the vector space, which is exactly a basis.

Let $A = GL_n(\mathbb{C})$ be diagonal, with distinct diagonal entries which are m th roots of unity for some $m \geq n$. Then consider the n -dimensional representation of $C_m = \langle g \rangle$ with $\rho(g) = A$. Then I claim the decomposition into subrepresentations is exactly $\mathbb{C}^n = \mathbb{C}e_1 \oplus \dots \oplus \mathbb{C}e_n$.

Let $G = C_3 = \{1, g, g^2\}$, and $\mathbb{C}[G]$ be the regular representation, then multiplication by g has matrix:

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & \zeta^2 & \zeta \\ 1 & \zeta & \zeta^2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \zeta & 0 \\ 0 & 0 & \zeta^2 \end{pmatrix} P^{-1}$$

so we are in the case of the previous example, and so this representation splits as the sum of three one-dimensional representations. This example can be generalised to any C_n , naturally

1.3 Schur's lemma

Theorem 1.3.1 (Schur's lemma). Let (V, ρ_V) and (W, ρ_W) be irreducible representations of G :

- (i) If $T : V \rightarrow W$ is a G -linear map, then either T is an isomorphism or the zero map.
- (ii) Suppose V is finite-dimensional, if $T : V \rightarrow V$ is G -linear, then $T = \lambda I$ for some $\lambda \in \mathbb{C}$.

Proof. (i) Apply the first isomorphism theorem, and consider the possibilities for $\ker(T)$ and $\text{im}(T)$

(ii) By linear algebra, every nonzero T is a nonzero eigenvector v such that $T(v) = \lambda v$, now $T - \lambda I$ is a non-injective map, so by the previous part $T = \lambda I$. $\square$